

Diverse invertebrate fauna using dry sediment as a refuge in semi-arid and temperate Australian rivers

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Abstract Dormant aquatic invertebrates can remain viable in riverbed sediment during dry phases, forming a source for recolonisation during wet periods. Regional differences in capacity for invertebrates to survive drying in this way are poorly understood, but may indicate regional differences in vulnerability to altered flow regimes. We compared diversity of invertebrates in dry sediment from intermittent rivers in temperate and semi-arid Australia after 4–8 weeks of drying. We predicted adaptations of semi-arid biota to severe and unpredictable drying would make dry sediment a more significant recolonisation source, with higher relative diversity when compared with temperate rivers. Emerging aquatic invertebrate assemblages were compared to those sampled in nearby pools, as a common drying refuge. Relative taxa richness in rehydrated sediments was higher in

the semi-arid region ($83 \pm 16\%$ of pool taxa) than the temperate ($47 \pm 6\%$ of pool taxa), despite lower overall richness (24 taxa in semi-arid, 32 taxa in temperate). Semi-arid rivers had greater potential for dry riverbeds to act as a source for recolonisation, given high relative diversity and abundance in dry sediment, combined with the frequent absence of alternative refuges. However, dry riverbeds in both regions provided a significant short-term refuge for aquatic invertebrates.

Keywords Dormancy · Climate · Resilience · Resistance · Macquarie River

Introduction

River systems that dry intermittently are common globally, making up the majority of rivers in Australia (Kennard et al., 2010) and other climate regions (Larned et al., 2010). In these rivers, the frequency and duration of drying is increasing, due to compounding effects of river regulation, abstraction and climate change (Kingsford, 2011; IPCC, 2013). Disruption of natural variability and failures to identify the extent of intermittent waterways threaten their unique biodiversity (Lake & Bond, 2007; Vörösmarty et al., 2010). Biota in intermittent rivers can account for a high proportion of regional biodiversity when spatial and temporal turnover are considered, with communities

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present often having little overlap with nearby perennial counterparts (Datry et al., 2014). It is therefore pressing to understand processes that maintain diverse biota of temporary waterways so they may be conserved (Acuña et al., 2014).

Aquatic invertebrates require physiological and behavioural adaptations to survive naturally variable patterns of drying and rewetting in intermittent rivers (Boulton, 1989; Stanley et al., 1994). Aerial dispersal and drift from refuge pools or perennial reaches are critical colonisation processes for recovery (Acuña et al., 2005; Sheldon et al., 2010; Chester & Robson, 2011; Datry et al., 2014). Pools and river waterholes are well-known ecological refuges, with pools identified as the most ‘popular’ refuge in intermittent rivers of Australia (Boulton et al., 2014). The dry river bed is also potentially a refuge and source for recolonisation, although less well understood. Invertebrates have adaptations for in situ survival, including production of desiccation-resistant eggs, cocoons and cysts, and diapause in larval and adult invertebrates (Strachan et al., 2015), forming an invertebrate ‘egg bank’ within dry river sediment (Brock et al., 2003). Resistance in dry sediment has been established as a highly productive survival mechanism in intermittent floodplain environments (Boulton & Lloyd, 1992; Tronstad et al., 2005; Jenkins & Boulton, 2007), and there is evidence it plays an important role in some river channels (Boulton et al., 1992; Larned et al., 2007; Datry et al., 2012; Storey & Quinn, 2013). Refuge availability is highly variable regionally, dependent on climate and geological characteristics, as well as spatial arrangement and barriers to dispersal in the landscape (Robson et al., 2013; Datry et al., 2014).

We know little of the specific requirements of dormant invertebrates across different habitats, where adaptations to drying vary as do duration and intensity of drying disturbances. Significance of the egg bank as a refuge and recolonisation source across different climate regions remains equivocal, but is better known in rivers in temperate or Mediterranean regions (Bogan et al., 2013), with only one arid river included in a recent global-scale meta-analysis (Stubbington & Datry, 2013). Much of current understanding comes from studies of single rivers or regions, with few comparative analyses until now (but see Boulton et al., 1992; Stubbington & Datry, 2013; Datry et al., 2014; Leigh et al., 2016).

Differences in responses of aquatic invertebrates to rehydration may reflect variations in sampling techniques, habitat and the phylogenetic origin of the fauna present. For example, aquatic biota in arid regions experience more frequent, severe and unpredictable drying (Marshall et al., 2006), and may be better adapted to drying disturbances than those in temperate regions (Boulton, 2003). Species in arid regions have life history, morphological or behavioural traits that enable them to survive unpredictable drying, such as diapause or asynchronous egg-hatching (Lytle & Poff, 2004). Their capacity for dispersal to refuges is probably also reduced given scarcity and isolation of water bodies amongst inhospitable habitat (Davis et al., 2013). Comparison of invertebrate responses to drying and rewetting across different climate regions increases our understanding of ecological adaptations to drying and the vulnerability of rivers to altered flow regimes.

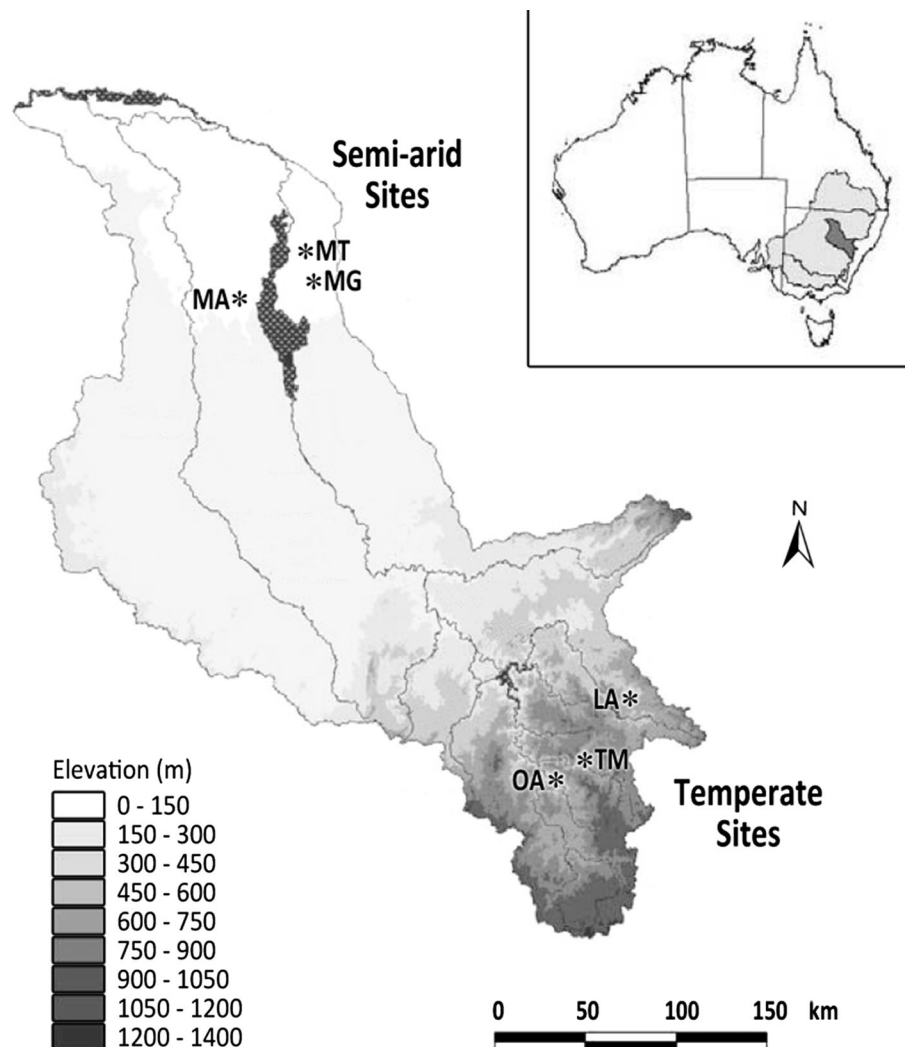
We investigated the significance of dry sediment as a short-term refuge for invertebrates and potential source for recolonization in rivers of the temperate and semi-arid regions in the Macquarie River catchment. We predicted that aquatic invertebrates inhabiting dry sediment after several weeks without flow would be more diverse in a semi-arid region, with higher taxa richness, functional diversity and abundance than those from temperate river sediment, relative to nearby refuge pools. We based these predictions on a hypothesis that there should be greater frequency of adaptations of semi-arid biota to severe and unpredictable drying, as well as reduced capacity for use of alternative refuges such as migration to pools or perennial reaches, compared to more mesic regions.

Materials and methods

Study sites and sampling

We sampled six rivers in the Macquarie River catchment (74,000 km²) of the Murray-Darling Basin in south-eastern Australia (Fig. 1), focusing on two climate regions for comparison (Table 1). The upper Macquarie River is temperate, receiving 600–1000 mm of annual rainfall, while the lower Macquarie River is a semi-arid region, receiving 300–500 mm of annual rainfall (CSIRO, 2008). Typically, the temperate rivers

Fig. 1 Macquarie River Catchment in the Murray-Darling Basin, South-eastern Australia, showing locations (*asterisks*) of river sections sampled in temperate and semi-arid regions: Marra Creek (MA), Marthaguy-Terrigal Creek (MT), Marthaguy Creek (MG), Lawsons Creek (LA), Oakey Creek (OA), Two-Mile Creek (TM)



dry seasonally in summer, while the semi-arid rivers dry unpredictably.

Within each climate region, nine sites were on intermittent rivers. Sites in the temperate region were low-order confined headwater tributary rivers (elevation 500–700 m), compared to semi-arid rivers comprising low-gradient floodplain tributaries (elevation 140 m, Fig. 1). All rivers had ceased to flow 30–62 days before sampling, between 17 and 19 January 2013 for temperate rivers, and 13–15 February 2013 for semi-arid rivers. Flow cessation was observed directly using time-lapse wildlife cameras deployed at each river site. Rivers formed a series of disconnected pools, with no flowing water and limited hyporheic connectivity, confirmed using a dye study. Temperate sites were generally sand and cobble-

based, underlain by bedrock; semi-arid sites were clay-based, overlain with sand in places. Rivers sampled had similar catchment-scale land use, comprising mixed shrubland and grasses with some grazing by cattle and kangaroos. We measured relative hydroperiod as the ratio of wetted observations to total observations for each site (see Larned et al., 2007; Jan 2012–April 2013, $n = 10$ for all sites) and days since flow as the number of days since the river had last flowed before sample collection.

We compared invertebrate assemblages inhabiting dry sediment in temperate and semi-arid regions, with reference to communities in nearby pools, a well-known refuge for a diverse range of river fauna. This allowed us to account for differing regional taxa richness and composition, providing local context for

Table 1 Regional habitat differences between semi-arid and temperate sites in the Macquarie River Catchment, south-eastern Australia

Category	Variable	Temperate region			Semi-arid region			Comparison	
		Mean	SD	<i>n</i>	Mean	SD	<i>n</i>	<i>t</i>	<i>P</i>
Climate	Köppen classification	Class C (temperate/mesothermal)			Class B (desert and semi-arid)			–	–
	Daily high temperature (°C) ^a	22.4	6.7	8567	26.4	7.4	8737	35.6	<0.001
	Summer daily high temp. (°C) ^a	27.4	4.5	2076	34.1	4.7	2166	32.7	<0.001
Hydrology/flow	Modelled runoff (mm)	60–200	–	–	5–20	–	–	–	–
	Hydroperiod (% time flowing)	65	9	30	56	20	30	–2.94	0.009
	Days since flow	53	14	3	40	13	3	–3.26	0.003
Physical characteristics	Elevation (m)	590	54	9	144	6	9	35.45	<0.001
	Slope of stream segment	0.76	0.45	9	0.03	0.01	9	5.78	0.002
	River shading (%)	65	14	9	59	8	9	–1.16	0.255
Substrate	Sediment moisture (%)	13.1	7.9	18	9.3	3.8	18	–1.73	0.096
	Moisture-holding capacity (%)	21.6	7.8	18	26.2	4.3	18	2.20	0.037
	Organic content (%)	4.4	3.1	18	3.9	2.2	18	–0.56	0.577
	Cobble (%)	10.5	16.4	18	0.0	0.0	18	–2.65	0.017
	Gravel (%)	37.8	15.0	18	1.9	1.6	18	–9.90	<0.001
	Sand (%)	32.3	10.0	18	43.6	19.8	18	2.13	0.043
	Fines (%)	19.3	13.3	18	54.1	20.4	18	5.93	<0.001
Water quality (refuge pools)	Temperature (°C)	25.4	3.2	18	26.7	2.3	18	–0.75	0.459
	Dissolved Oxygen (mg L ^{–1})	4.2	2.2	18	4.6	6.6	18	0.02	0.982
	pH	7.6	0.5	18	8.2	0.8	18	–2.57	0.019
	Turbidity (NTU)	45	18	18	173	178	18	–2.37	0.035
	Conductivity (µs cm ^{–1})	1303	321	18	654	93	18	7.49	<0.001
Water quality (sediment microcosms)	Temperature (°C)	25.8	3.4	72	27.2	4.3	72	–1.81	0.083
	Dissolved Oxygen (mg L ^{–1})	7.9	1.8	72	9.3	1.1	72	4.15	<0.001
	pH	8.3	0.5	72	8.0	0.4	72	–2.38	0.023
	Turbidity (NTU)	93	130	72	228	183	72	4.68	<0.001
	Conductivity (µs cm ^{–1})	1039	536	72	575	279	72	–6.23	<0.001

All environmental variables measured at river scale or below included in model selection. Modelled runoff from CSIRO (2008), Köppen classification from Peel et al. (2007), elevation and slope derived from Stein et al. (2012), all other variables measured in the current study unless specified

^a Australian Bureau of Meteorology: <http://www.bom.gov.au>; accessed May 2016, daily temperature data 1990–2015, Mudgee and Nyngan Airport stations used for Temperate and Semi-arid regions respectively

variation to determine relative importance of dry sediment as a refuge in each climate region. We used a nested sampling design, choosing three rivers in each region, and then randomly selecting one typical and geomorphically distinct section (600–3000 m) within each, with variable lengths necessary to encompass at

least two refuge pools under no-flow conditions. Three dry sites (100 m length) were chosen between pools on each river. Pools were sampled at the same time as dry refuges, providing a reference for overall aquatic assemblages present in each river. We took two composite replicate samples (0.024 m² × 4 samples

per composite) at each dry sediment site over a 100 m reach of the thalweg, using a spade. These were taken to a depth of 100 mm, given few invertebrates occur below this depth (Larned et al., 2007). In a pilot study, 92% of invertebrates were found at 0–100 mm compared with 100–200 mm, with no new taxa found in the deeper substrate. Samples were deposited in plastic aquaria (280 × 380 × 200 mm depth) and transported to the laboratory, limiting disturbance. We also collected one composite sediment sample (50 g × 4 wet mass) from each sampling site, transferred to sealed jars for analyses of moisture and organic content and grain size. We also randomly collected two composite invertebrate samples in each nearby refuge pool (0.07 m² × 6 samples per composite), using a flow-independent suction sampler (Boulton, 1985). We also measured dissolved oxygen, conductivity, temperature, pH and turbidity, using an Aquaread AM200 multi-parameter water quality metre prior to sampling refuge pools.

Laboratory assays and analyses

We rehydrated the dry sediment samples in microcosms in conjoined temperature controlled glass-houses, with ambient temperature ranges, typical of the two climatic regions (18–24°C for temperate samples and 18–28°C for semi-arid samples). Microcosms were aerated, and covered with mesh lids to retain natural photoperiod and prevent external invertebrate colonisation. Four control microcosms were filled with sterile sand. We flooded microcosms with deionised water (Jenkins & Boulton, 2003), and sampled invertebrates at 0, 7, 14 and 28 days after flooding, ceasing because pilot analyses showed deterioration in water quality and excessive algal growth after this. On Day 0, samples were collected 2 h after rehydration. We sampled invertebrates within microcosms by gently agitating the sediment and immediately syphoning all surface water through a 55 µm sieve. Invertebrates were live-counted using a Leica M125 stereo microscope in Bogorov zooplankton trays. They were identified to genus for insect and molluscan taxa, family for other arthropods, oligochaetes, turbellarians, hirudineans and platyhelminths. Ostracods and nematodes were not identified beyond class or phylum, respectively. Semi-aquatic taxa were included and terrestrial taxa excluded from analyses (see Online Resource 1; Table S1 for taxa list). Only a

single rotifer emerged from control microcosms. Rotifers were excluded from analyses because their small size meant they were not sampled reliably. Samples were replaced in microcosms immediately following identification to minimise loss of eggs, allow maturation of immature invertebrates and provide a complete assessment of the assemblage at the end of the rehydration period (Boulton & Lloyd, 1992). We also sampled dissolved oxygen, conductivity, temperature, pH and turbidity in microcosms on each sampling day.

We estimated organic content and grain size of dried sediment. First, we weighed and dried sediment samples at 105°C to constant mass (24–48 h) and then reweighed them, calculating the moisture content as the difference between initial and dry mass. About 50 g of each dried sample was combusted for 4 h at 400°C and reweighed, to estimate organic content from loss on ignition. The remaining dried sediment was used to measure grain size, according to ASTM Standard *D 6913–04* (ASTM, 2009). We soaked samples overnight in 10% sodium hexametaphosphate, dispersed using an ultrasonic water bath, and decanted through a 63 µm washing sieve. Sieve contents were dried overnight at 105°C, and shaken on a mechanical shaker for 10 min through nested sieves: 20, 6, 2 mm, 600, 200, 63 µm and pan. Masses were estimated for four categories: cobble (>20 mm), gravel (2–20 mm), sand (63 µm–2 mm) and silt/clay (<63 µm). Total mass was within 5% of the original for all samples.

Statistical analyses

We calculated three response variables: (1) taxa richness, (2) functional diversity, and (3) abundance of aquatic invertebrates. Taxa richness was the accumulated number of unique taxa over the four sampling periods for each sample. Abundance was the greatest number of each taxon on any of the four sampling days (0, 7, 14, 28), removing the possibility of non-independence caused by recounting the same individuals. Functional diversity, the range of functional traits present in a community (Petchey & Gaston, 2006), was used to account for similarities in the function of invertebrate assemblages produced by different taxa occupying similar niches. We identified 63 categories of 11 invertebrate traits to genus level where possible for functional diversity

(Usseglio-Polatera et al., 2000), reflecting a broad range of biological characteristics. These represented life history, feeding habit, dispersal and respiration traits (see Table S2 for full list). Taxa were assigned to more than one category where there was evidence of more than one mode used by an individual taxon for a specific trait, or where there was evidence of differences within a genus. This also allowed us to use multiple modalities where necessary for taxa that we had identified at a coarser level than genus. Functional diversity was calculated using the Functional Dispersion (FDis) metric in the FD R package (Laliberté et al., 2014). This estimated the amount of trait space occupied by the assemblage weighted by their relative abundances, based on Principal Component Analysis for each sample in multivariate trait space, and is independent of taxa richness (Laliberté & Legendre, 2010).

We tested the effect of climate region (temperate, arid) relative to pool assemblages using the interaction between climate region and habitat type (pools, dry sediment). Generalised Linear Mixed effect Models (GLMM) allowed for the unbalanced design between habitats and were performed using the lme4 R package (Bates et al., 2014). An offset in the models was used for abundance to account for variable sample sizes; abundances were converted to individuals per square metre for visualisation, however. Poisson error distribution was used for taxa richness, Gaussian error distribution for functional diversity and negative binomial distribution for abundance, based on the type of response and confirmed using examination of standardised residuals versus fitted values (Zuur et al., 2009).

We used 17 environmental variables as covariates for model selection for an additional set of models using only the dry sediment samples, i.e. not including pool samples, as these could not have the same set of environmental variables measured (Table 1). These were shortlisted to nine variables for climate region analysis by removing highly collinear variables, and any highly correlated with climate region. Wherever two explanatory variables were highly correlated (Pearson's $r > 0.7$), the variable with the weakest relationship with the response variables was removed (Dormann et al., 2013). All variables were scaled and normalised. Model selection was undertaken by removing environmental variables stepwise and comparing with the null model to find the best model for

each response variable, based on Akaike Information Criterion (AIC). The final two candidate models were compared using a Chi squared test, and the simplest model kept where there was no significant difference between models. We provided estimates of conditional R^2 (Nakagawa et al., 2013) for variation explained by each model, except those with negative binomial distribution.

To visualise the effect of climate region on multivariate assemblages of aquatic invertebrates, we used a non-metric multidimensional scaling (NMDS) ordination with Bray–Curtis similarity in the Vegan R package (Oksanen et al., 2013). Statistical tests of the effect of climate region were performed using a model-based analysis of multivariate abundance data (manyglm) in the mvabund R package (Wang et al., 2012). Mvabund is a multivariate implementation of generalised linear models with each taxon fitted to a separate generalised linear model (GLM), and row-resampling for multivariate inference accounting for correlation between taxa. We investigated variation in multivariate abundance between climate regions using the negative binomial distribution with rivers included as a fixed variable. The interaction between climate region and habitat was used to test the effect of climate region relative to pools on the same river. We examined responses of individual taxa with post hoc tests, using conservative adjusted P-values. These tests were superior to distance-based alternatives which heavily weight more abundant species, given there was a significant mean–variance relationship (Warton et al., 2012). Means were reported as mean $\pm$ standard error. All analyses were performed in R version 3.1.2 (R Core Team, 2014).

Results

The two climate regions generally had similar hydrology and water quality, with substrate and physical characteristics showing the most difference (Table 1). Temperate region sites had a slightly longer period between ceasing to flow and sediment sampling compared to semi-arid sites ($t = -3.26$, $P = 0.003$, Table 2). Substrate was coarser at temperate sites, dominated by gravel and sand with some cobble and fines. By contrast, substrate at semi-arid sites was almost entirely sand and fines (Table 1). Sediment

Table 2 Summary of model results for effect of climate region (temperate, semi-arid) and habitat (rehydrated sediment, refuge pools)

Response variable	Predictor variable	Parameter estimate	<i>z/t</i> value	<i>P</i> value	Model <i>R</i> ²
Taxa richness	Intercept	2.08	22.50	<0.001	0.59
	Habitat (wet)	0.33	2.26	0.024	
	Region (temperate)	0.37	3.07	0.002	
	Habitat: region	0.40	2.18	0.029	
Functional Diversity	Intercept	3.20	5.35	0.003	0.50
	Habitat (wet)	3.70	4.70	<0.001	
	Region (temperate)	0.82	0.97	0.376	
	Habitat: region	−0.91	−0.82	0.416	
Abundance	Intercept	5.75	23.09	<0.001	NA
	Habitat (wet)	−2.35	−4.73	<0.001	
	Region (temperate)	0.40	1.15	0.258	
	Habitat : region	1.78	2.53	0.015	

Best model for each response variable (taxa richness, functional diversity, abundance) from model selection process based on Akaike Information Criterion (AIC)

moisture, organic content and shading were similar between regions (Table 1). Temperature and dissolved oxygen were similar between regions, with higher conductivity ($t = 7.49$, $P < 0.001$, Table 1) and lower turbidity ($t = -2.37$, $P = 0.035$, Table 1) at temperate sites. Turbidity was particularly variable between sites (Table 1). Except for dissolved oxygen, water quality was similar between refuge pools measured in the field, and rehydrated sediment microcosms measured in the laboratory (Table 1).

As predicted, there were regional differences in invertebrate diversity between rehydrated sediment samples. Taxa richness was higher in semi-arid than temperate rehydrated sediment samples when compared with pool samples, indicated by a significant interaction between region and pooled or dry habitat ($z = 2.18$, $P = 0.029$; model $R^2 = 0.59$, Table 2). This is despite raw taxa richness being significantly higher in temperate than semi-arid samples ($z = 3.07$, $P = 0.002$, Table 2). Forty-three aquatic invertebrate taxa emerged from dry sediment samples ($n = 36$ composites) in total. (Online Resource 1). None of the environmental variables significantly explained taxa richness differences between regions (Table 3). Comparative functional diversity was the same between regions, both comparative to pools ($t = -0.82$, $P = 0.416$; model $R^2 = 0.50$, Table 2) and using raw data ($t = 0.97$, $P = 0.376$, Table 2). Turbidity, sediment moisture and organic content explained

much of the observed variation between samples (model $R^2 = 0.78$, Table 3). Semi-arid sites had higher invertebrate abundance than temperate sites when compared with equivalent pool abundance, indicated by a significant interaction between region and habitat ($t = 2.53$, $P = 0.015$, Table 2). There were more than 30 times as many invertebrates in dry sediment compared with nearby pools in the semi-arid rivers, compared with 7 times as many in temperate sediments as in nearby pools (Fig. 2). Raw invertebrate abundance was similar between climate regions ($t = 1.15$, $P = 0.258$, Table 2), but highly variable between samples (691 ± 251 individuals per m^2 temperate, 850 ± 693 individuals per m^2 semi-arid; Fig. 2). None of the environmental variables contributed to explaining abundance differences between samples (Table 3).

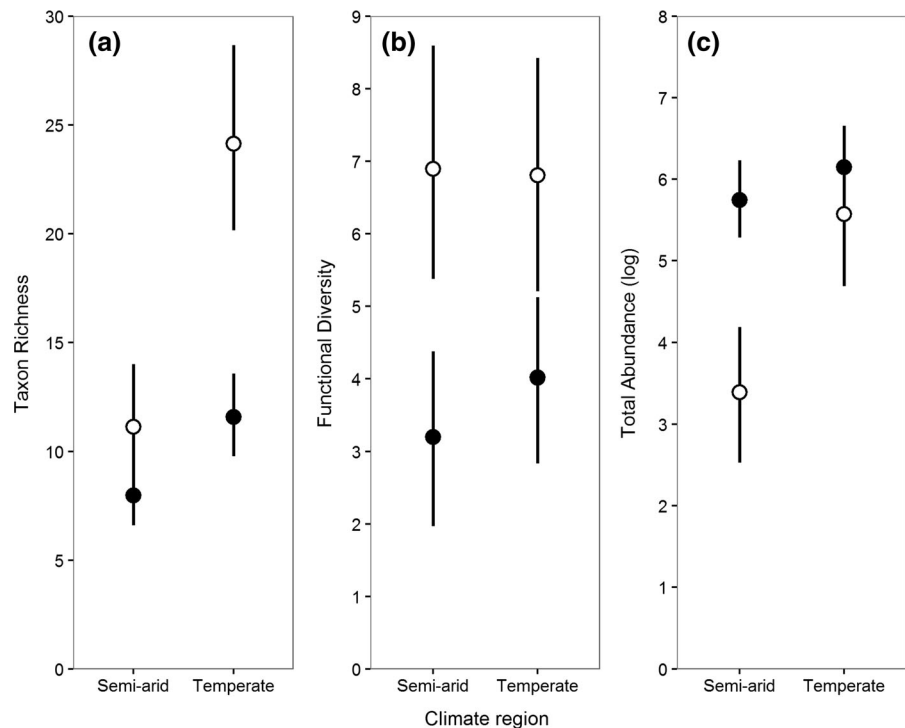
Aquatic invertebrate assemblages were taxonomically distinct between climate regions ($t = 10.34$, $P < 0.001$; Fig. 3). Lack of significant interaction between climate region and habitat indicates that both temperate and semi-arid regions show a similar shift in assemblage from pools to dry riverbed ($t = 2.37$, $P = 0.423$; Fig. 3). Moisture-holding capacity of the sediment ($t = 9.07$, $P = 0.016$), and number of days since the creek was last flowing ($t = 12.73$, $P = 0.012$) were also significant in the best model. Regional differences were driven by leptophlebiid mayfly taxa (*Atalophlebia* sp. and *Tasmanocoenis* sp.)

Table 3 Summary of model results for environmental factors affecting aquatic invertebrate assemblages emerging from rehydrated sediment

Response variable	Predictor variable	Parameter estimate	<i>z/t</i> value	<i>P</i> value	Model <i>R</i> ²
Taxa richness	Intercept	2.08	24.95	<0.001	0.27
	Region (temperate)	0.37	3.44	0.005	
Functional diversity	Intercept	3.39	4.29	<0.001	0.78
	Region (temperate)	−2.00	−1.79	0.074	
	Turbidity	−1.61	−3.19	0.001	
	Sediment moisture	1.14	2.75	0.006	
	Organic content	−1.04	−2.58	0.010	
Abundance	Intercept	6.06	5.16	<0.001	NA
	Region (temperate)	0.63	0.38	0.703	

Best model for each response variable (taxa richness, functional diversity, abundance) from model selection process based on Akaike Information Criterion (AIC), starting with 17 environmental variables measured for dry sediment samples (Table 1)

Fig. 2 Mean $\pm$ 95% confidence intervals using predicted values from models for **a** taxonomic richness, **b** functional diversity and **c** total abundance of invertebrates in rehydrated sediment (filled circles) and refuge pools (open circles), from semi-arid and temperate rivers in South-eastern Australia



that dominated temperate sites (Fig. 4), contributing around one third of the observed variation between regions (Fig. 5). Microcrustacean taxa were dominant at semi-arid sites (Fig. 4), with Moinidae, Macrothricidae and Calanoida contributing most to differences (Fig. 5). Many taxa were common within rehydrated sediment from one region and entirely absent from the other, with 29 of 43 taxa recorded as present in one

region only (Online Resource 1). In the semi-arid region, these were mostly crustaceans (*Branchinella* sp., *Triops* sp., *Caenestheriella* sp, Macrothricidae and calanoid copepods), with some insect taxa (*Aedes* sp.) and Acarina (Mesostigmata). Several insect orders (Hemiptera, Odonata and Coleoptera) were only found at temperate region sites, as well as Ephemeroptera (Caenidae and Leptophlebiidae), Diptera (Tipulidae

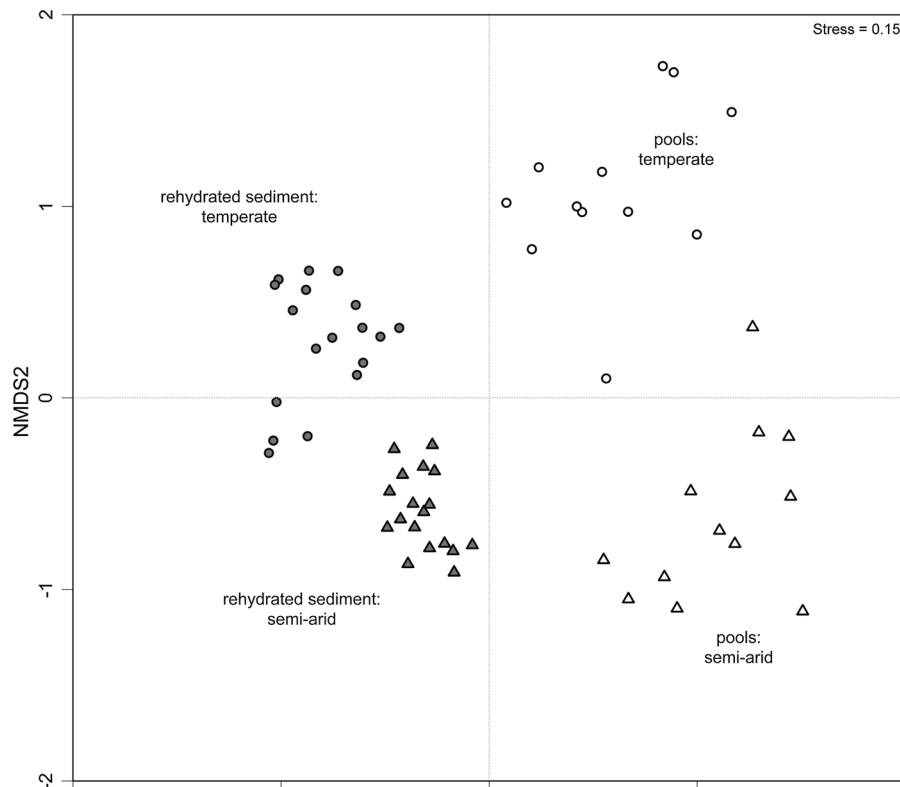


Fig. 3 Similarity between invertebrate assemblages from rehydrated sediment (filled symbols) and pools (open symbols) in semi-arid (triangles) and temperate regions (circles), using

non-metric multidimensional scaling (NMDS) for all samples, based on Bray–Curtis similarity

Fig. 4 Invertebrate taxa associated with dry sediment in temperate and semi-arid regions, size of text represents frequency of occurrence (number of sites, $n = 36$), position on x axis represents frequency at temperate versus semi-arid sites for each taxon, with taxa centred on vertical dotted line present equally in both regions. All taxa recorded in dry sediment shown, underlined taxa indicate significant difference between regions from multivariate model (manyGLM). Taxa marked with an asterisk are early instars

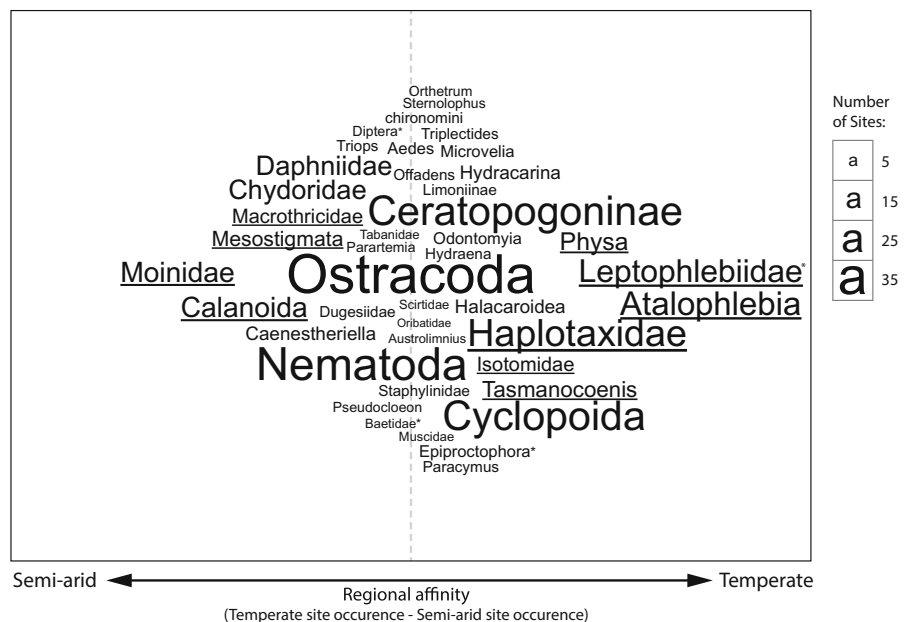
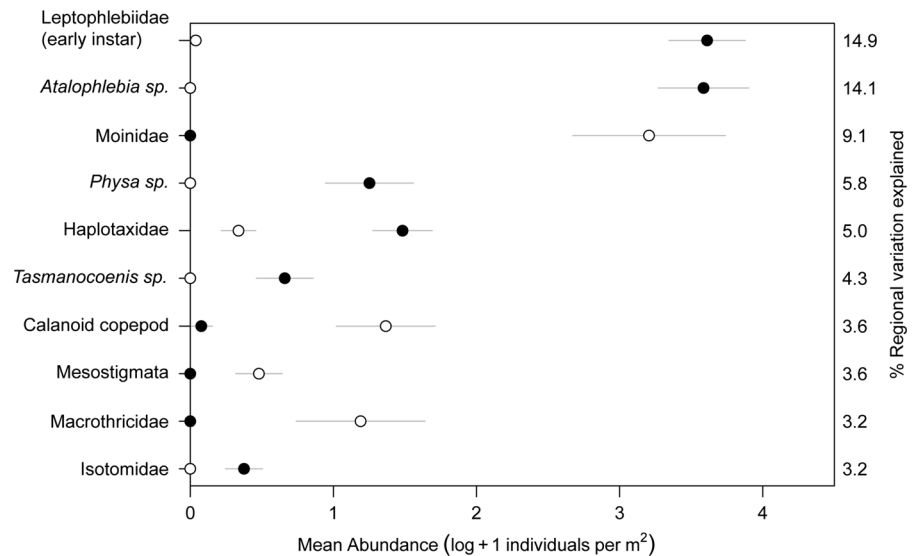


Fig. 5 Mean abundance and 95% confidence intervals of significant taxa ($P < 0.05$) contributing to multivariate assemblage difference between temperate (filled circles) and semi-arid dry river sediments (open circles), ordered by proportion of total contribution to between- region variation



and Stratiomyidae), Mollusca and Acarina (Halacaridae and Hydracarina). Ostracods were ubiquitous, recorded from every sample across both regions.

Assemblages in rehydrated sediment were distinct from those in pools ($t = 10.27$, $P = 0.001$), with rehydrated sediment assemblages more like one another between samples than variable pool assemblages (Fig. 3). Several taxa recorded in refuge pools were not present in rehydrated sediment: Decapoda (*Paratya* sp., *Macrobrachium* sp., *Cherax* sp.) and Hemiptera. Conversely, several taxa were consistently unique to rehydrated sediment, representing Crustacea (*Branchinella* sp., *Triops* sp., *Caenestheriella* sp.), Acarina (Halacaridae and Mesostigmata), Diptera (*Aedes* sp. and Tipulidae) and Dugesidae (Online Resource 1).

Discussion

Dry sediment refuges have been known for centuries, although their significance was underestimated historically (Strachan et al., 2015). Their critical role in recolonisation following drying in rivers is increasingly understood (Williams, 1977; Stubbington & Datry, 2013). We identified the significance of rehydrated sediment as a potential source of recolonising invertebrates in semi-arid rivers, reflecting adaptations of semi-arid invertebrate assemblages to drying disturbance. Dry sediment from temperate rivers also had a diverse invertebrate fauna, with at least half of the

taxa from nearby pools emerging from rehydrated sediment (Fig. 2). Historically, invertebrates in dry sediment were considered specialists, making a poor contribution to recolonization compared with resilience mechanisms such as aerial colonisation and drift (Stanley et al., 1994; Fritz & Dodds, 2004; Chester & Robson, 2011). However, this study shows dry sediment to be a significant refuge promoting diverse invertebrate assemblages in semi-arid and temperate rivers, albeit with both similarities and differences between regions.

Differences between climate regions

Dry sediments from semi-arid rivers had a higher proportion of taxa from adjacent refuge pools as predicted, but lower raw taxa richness than temperate rivers (Fig. 2; Table 2). Our semi-arid fauna was more diverse than that from other semi-arid regions, which could be related to substrate (Chester & Robson, 2011) and desiccation differences (Stanley et al., 1994), as well as differences in taxa groups assessed, and a relatively short drying time. Substrate differences between regions in the current study likely contributed to some observed faunal differences. Meiofaunal dominance in the semi-arid region contributed to high observed taxa richness and abundance. Microcrustacea (Cladocera, Copepoda and Ostracoda) were dominant in the semi-arid region, reflecting high abundance following flooding (Jenkins & Boulton, 2003; Boulton et al. 2006). Comparable rehydration studies often

focus on macroinvertebrate taxa only (Stubbington & Datry, 2013). Semi-arid habitats without a significant microcrustacean fauna have low diversity in rehydrated sediment, such as headwater streams lacking fine sediment, or where the phylogenetic origin of the fauna present was wetter (Chester & Robson, 2011).

Several taxonomic assemblage differences were observed between this study and the majority of semi-arid studies that rehydrated sediments after longer drying timeframes (Boulton & Lloyd, 1992; Jenkins & Boulton, 2003). A range of insect taxa were recorded in the semi-arid region. These were likely to be resistant to short-term drying only, relying on aestivation rather than emerging from desiccation-resistant stages. Mayflies particularly are not known to use dry sediment as a refuge in drier climates (Robson et al., 2011). Insects are also generally more sensitive to desiccation than microcrustaceans (Paltridge et al., 1997); therefore, microcrustacean-dominated assemblages at semi-arid sites may be more resistant to long term drying compared with temperate sites. Although prolonged drying was not investigated, this would be an interesting area for further research. While refuge pools were present in the semi-arid region at the time of sampling, scarcity of persistent pools in low-gradient arid rivers means differences between climate regions are likely to be more pronounced during extended drying. Early successional and temporary pool specialist species (e.g. Anostraca, Notostraca and Spinicaudata) in the semi-arid zone only are indicative that this region is potentially more resistance-dominated in terms of recolonisation (Datry et al., 2014). These taxa are likely to be particularly vulnerable to the effects of managed flows often encountered in intermittent rivers (Hughes, 2005).

Functional diversity was similar between semi-arid sediments and temperate sites (Fig. 2; Table 2). Given higher raw taxa richness at temperate sites, this infers a relatively higher level of redundancy, with multiple taxa performing the same functional roles. Limited trait information may have limited the discriminatory power of functional diversity measures, however, particularly where a variety of strategies are used by taxa within families for example (Robson et al., 2011). Overall, similar functional diversity between climate regions, indicates no added resilience to disturbance is given by the additional taxa in temperate rivers (Walker, 1992; Allison & Martiny, 2008).

Within 28 days of rehydration, total abundance of invertebrates was higher than pool abundance in semi-arid rivers but not temperate rivers, consistent with predictions (Fig. 2; Table 2). This reflects a ‘boom’ in productivity in the semi-arid region following flooding (Boulton & Lloyd, 1992; Jenkins & Boulton, 2003). Abundance contrasts between single-region studies are difficult, as abundance measurement is more sensitive to differences in sampling and rehydration methods than taxa richness, for example. Additionally, exponential decline in invertebrate numbers following water loss may partially account for high variability (Larned et al., 2007). Stanley et al. (1994) found abundances up to ten times higher in a semi-arid region; however, numbers declined rapidly over a few weeks after water loss.

Factors driving observed differences between regions were equivocal. Regional differences in taxa richness have been attributed to reduced moisture levels and increased drying severity, including both high temperatures and speed of drying (Boulton et al., 1992). Only functional diversity and multivariate assemblage difference were correlated with moisture in the current study (Table 3), contrasting with observations that increasing moisture improves invertebrate viability (Stanley et al., 1994) and increases taxa richness and abundance (Stubbington & Datry, 2013). Sediment moisture could become critical to invertebrate seedbank viability over a drying period longer than the several weeks of our study (Jenkins & Boulton, 2007; Stubbington et al., 2016). Undoubtedly, the relationship between sediment moisture, preceding dry periods and invertebrate assemblages remains complex (Larned et al., 2007; Datry et al., 2012). High environmental harshness (based on climate and moisture) has been correlated with a lower proportion of macroinvertebrates (Stubbington & Datry, 2013), although arid climates were under-represented in this synthesis. Conversely, recent evidence suggests arid regions have a higher proportion of taxa using dry sediment as a refuge than temperate or Mediterranean climates, similarly to the present study (Leigh et al., 2016). Invertebrate assemblages adapted to dry conditions of semi-arid rivers may be more likely to use dry sediment as a refuge, given the rarity of perennial reaches or pools in arid regions (Davis et al., 2013).

Similarities between climate regions

Taxa richness in dry sediments was high in both regions following rehydration, compared to many microcosm experiments (Stanley et al., 1994; Larned et al., 2007; Chester & Robson, 2011), although similar to some short-term drying studies (Boulton, 1989; Tronstad et al., 2005; Datry et al., 2012). Most rehydration studies have 50% or less of the taxa from flowing rivers in dry sediment (Stubbington & Datry, 2013), compared to our greater than 50% from nearby pools in both temperate and semi-arid rivers (Fig. 2).

Compositionally, our invertebrate assemblages in both regions were similar to others from dry sediments (Boulton, 1989; Tronstad et al., 2005; Datry et al., 2012), including common taxa (microcrustacea, Ephemeroptera, Nematoda and Ceratopogonidae). There were some relative differences; Ephemeroptera (mayflies) were generally more abundant and chironomids were rare. Mayflies are generally reliant on perennial water (Strachan et al., 2015), but genera recorded were specialised to drying, with Baetidae and Caenidae known to hatch from desiccation-resistant eggs (Paltridge et al., 1997) and *Atalophlebia* sp. common in receding pools (Boulton, 1989). Only a few chironomid individuals (early instar Chironomini) occurred in rehydrated sediment, compared to at least 17 genera from adjacent refuge pools. There was no obvious explanation for their scarcity, given their larvae commonly emerge during inundation experiments (Boulton, 1989; Tronstad et al., 2005; Larned et al., 2007; Datry et al., 2012). However, specific cues for chironomid emergence from sediment, such as temperature and photoperiod, may have been lacking in this study, with chironomid life histories highly seasonal (Armitage et al., 1995; Tronstad et al., 2007).

Interestingly, several aquatic taxa (Branchiopoda and Diptera) not present in refuge pools emerged from rehydrated sediments, reflecting known exclusivity to dry sediment (Boulton & Lloyd, 1992). Several semi-aquatic and terrestrial taxa (Acari, Collembola, Staphylinidae) also only occurred in rehydrated sediment samples, reinforcing the importance of dry riverbeds as habitats (Steward et al., 2011). Conversely, Decapoda and Hemiptera in refuge pools were not recorded in rehydrated sediment. Only *Cherax* sp. (Decapoda) can resist desiccation but was probably not sampled in our study because it burrows at the river's edge (Williams, 1980).

By comparison with pools, dry sediment in both regions may provide a more stable refuge for specialist taxa. Conditions altered rapidly in pools following their isolation, contributing to deteriorating water quality including high temperatures (Boulton, 1989; Sheldon & Fellows, 2010), overcrowding and isolation of predators (Lake, 2000; Covich et al., 2003). These factors probably contributed to the high taxonomic variability of assemblages among pool samples compared to our rehydrated sediment samples (Fig. 3).

Microcosm rehydration

Our short-term microcosms probably approximated the function of inundated riverbeds reasonably well (Jenkins & Boulton, 2003). Taxa richness provided a reliable estimate of the invertebrate assemblage because it only relies on a few individuals of each taxon emerging from rehydrated sediment. Some taxa requiring seasonal cues or chemical cues from river water were probably not represented (Brock et al., 2003). Further, deep-burrowing taxa may have been inadequately sampled. Dormant taxa such as chironomids, which can take more than a month to emerge (Tronstad et al., 2005), were also possibly under-represented. We did not sample hyporheic zones beneath river channels, nearby persistent perennial reaches, and off-river habitats such as ponds, dams and springs that may have contributed to differences in invertebrate assemblages (Boulton et al., 1992; Chester & Robson, 2013). Closed microcosms enhance some ecological processes (e.g. predation) and exclude others such as nutrient inputs and colonisation (Boulton & Lloyd, 1992). Intense biotic interactions in our closed systems and patchily distributed sampling habitat may have increased variation (Giri et al., 2010). Despite these drawbacks, similar microcosms realistically reflected abundance estimates in extant waterbodies (Jenkins & Boulton, 2007). Importantly, microcosms allowed comparative assessment across similar studies, particularly as opportunistic studies of drying and flooding are inherently difficult in situ (see Jenkins & Boulton, 2003).

Conclusions

The surprisingly high taxonomic richness of invertebrate assemblages of rehydrated dry sediment, along

with rapid emergence of abundant invertebrates indicated considerable recolonisation potential. Dry sediment refuges have the potential to substantially contribute to high productivity ‘boom’ conditions following drying, even without recolonisers via drift or aerial dispersal. Dry sections of intermittent rivers should be considered important to maximising diversity, with maintenance of wetting and drying patterns contributing to beta-diversity (Datry et al., 2014). Increased dry periods (Boulton & Lloyd, 1992), or increased flow duration may decrease viability of invertebrate assemblages in intermittent rivers.

The high proportion of invertebrate taxa using dry sediment and presence of temporary habitat specialists in the semi-arid region means dry sediment may be particularly important to viability of diverse invertebrate populations, especially given frequent absence of refuge pools. Abundances from rehydrated sediment in semi-arid rivers were higher than those in remaining refuge pools, within 1–4 weeks. The semi-arid invertebrate fauna appears better adapted to drying than its temperate counterpart, making the semi-arid taxa potentially more tolerant of increased drying imposed by human demand and climate change.

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